CSE 29 Week 4 Discussion

Hashing, Project 2, Stacks, Dynamic Memory Allocation and more

Announcements

- Project 2 due **10PM** this Friday, 10/23
- Project 1 resubmission due **10PM** Friday, 10/23
- Lab 4 check-off due **11AM** Monday, 10/26

Hashing review

- Hash Function = a one-way function that maps inputs of arbitrary size into fixed-sized values
- E.g: SHA256 maps from X -> 256 bit binary string
- Some Important Properties:
  - Deterministic
  - One-way
  - Collision Resistant

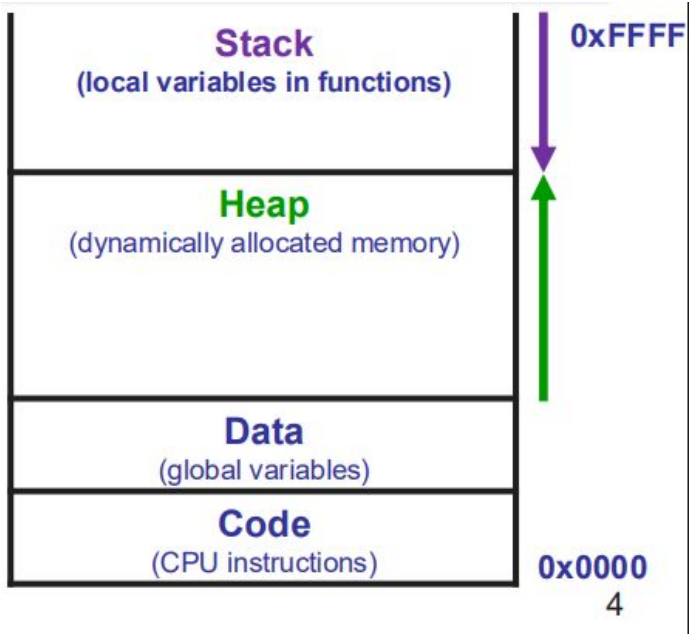
Project 2

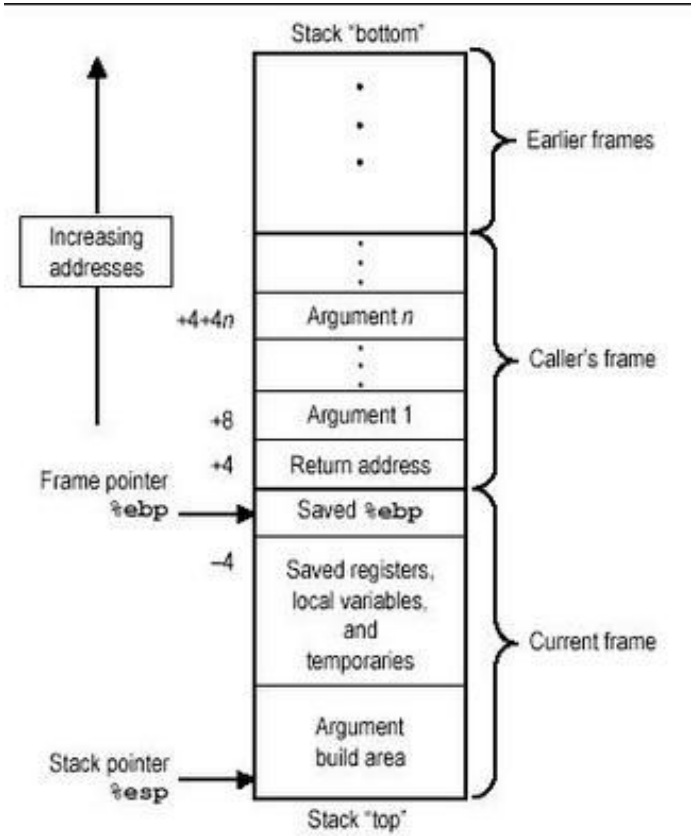
PA2 - Hashing and Passwords: Due Friday 10/23 at 10:00pm

Cryptographic hash functions take an input of arbitrary length and produces a fixed length output. The special features are that the outputs are *deterministic* (the same input always generates the same output) and yet the outputs are apparently "random" – two similar inputs are likely to produce very different outputs, and it's difficult to determine the input by looking at the ouput.

A common application of hashing is in storing passwords. Many systems store only the hash of a user's password along with their username. This ensures that even if someone gains access to the stored data, users' actual passwords are not exposed. When a user types in a password on such a system, the password handling software grants access if the hash value generated from the user's entry matches the hash stored in the password database.

Stack Frame


Stack
(local variables in functions)
Heap
(dynamically allocated memory)
Data
(global variables)
Code
(CPU instructions)
0xFFFF
0x0000


Stack "bottom"
Increasing addresses
+4+4n
Argument n
+8
Argument 1
+4
Return address
Frame pointer %ebp
Saved %ebp
–4
Saved registers, local variables, and temporaries
Argument build area
Stack pointer %esp
Stack "top"
Earlier frames
Caller's frame
Current frame

Questions

- **Project 2**
- Lecture content: pointers, hashing, stacks/heaps, ...
- Labs: unix commands, git, vim, ...
- etc